



Product Requirement Document (PRD)

Dividend Compass V2.0

The Bidirectional Dividend Planning Platform

DOCUMENT METADATA

Author : Jason Chandra
Role : Technical Product Manager / Business Analyst
Version : 2.0 (Multi-Tab Bidirectional Platform)
Tech Stack : Python, Streamlit, yfinance API
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REVISION HISTORY

Version	Date	Description of Change	Author
v1.0	May 15, 2026	Initial draft specification for the Forward Lifestyle Calculator engine.	Jason Chandra
v1.1	May 22, 2026	UI/UX Refinement: Optimized user input flow, enhanced layout aesthetics for better accessibility, and streamlined forward calculation rendering.	Jason Chandra
v2.0	June 8, 2026	Upgraded to multi-tab bidirectional framework. Added Reverse Goal Planner core math logic, data dictionary, and exception handling mapping.	Jason Chandra

1. Executive Summary & Product Vision

Dividend Compass is a serverless, database-less financial technology application designed to simplify asset management and investment planning for the Indonesian stock market (IDX). The platform removes the traditional overhead of complex financial spreadsheets by providing a highly optimized, dual-engine planning system wrapped in a minimalist, dark-themed user interface.

In its V2.0 release, the platform transitions from a single-threaded forward calculator into a sophisticated **bidirectional simulation platform**. By leveraging real-time data ingestion and programmatic sanitization, Dividend Compass enables retail investors to map out investment journeys from two separate starting points: asset volume (Lifestyle Calculator) and financial targeting (Reverse Goal Planner).

2. System Architecture & Component Responsibilities

The system follows a lightweight, serverless three-tier architecture that guarantees high performance, security, and low maintenance costs by shifting state management directly to the client session.

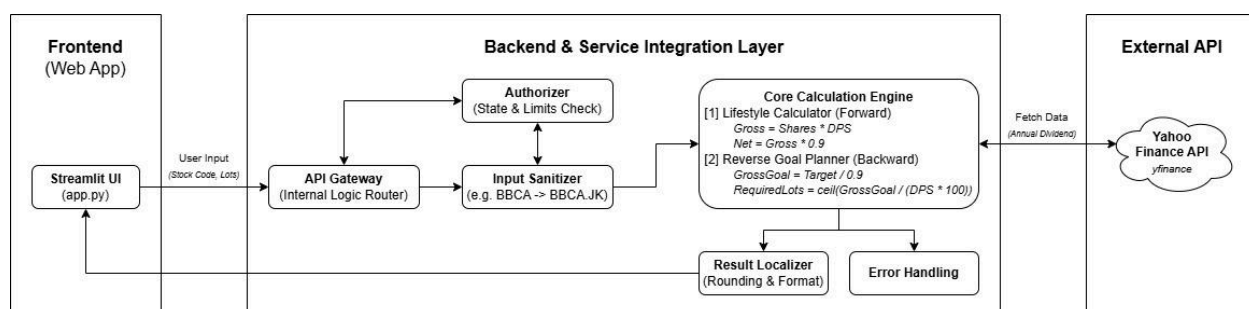


Figure 2.1 System Architecture

Component	Responsibility	Technology / Tool
User Interface (UI)	Manages the multi-tab state, implements the custom minimalist dark theme CSS variables, captures user inputs, and handles contextual result rendering.	Streamlit (Python) & Custom CSS Injection
API Gateway / Router	Intercepts frontend requests, validates query parameter constraints, and routes the internal	Internal Logic Router (Python)

Component	Responsibility	Technology / Tool
	logic flow between forward and backward engines based on active tab state.	
Input Sanitizer	Normalizes user inputs. Automatically sanitizes IDX stock tickers by standardizing letters and appending the .JK suffix to comply with regional data routers.	String Manipulation Module
Data Integration Layer	Executes non-blocking upstream network requests to fetch real-time market close prices and historical annual Dividend Per Share (DPS) distributions.	yfinance (External API Engine)
Core Logic Engine	Executes bidirectional math models: translates share portfolios into net passive income, or reverse-engineers revenue milestones into required lot acquisitions.	Python Math Logic
Localization & Format Service	Applies regional accounting standards to numerical results, handling thousand separators (dots), currency indicators (Rp), and rounding standards.	Python Localizer Formatting
Error Handler & Safe Boundary	Prevents system crashes by providing safe try-except boundaries for invalid tickers, connection timeouts, or zero-dividend asset errors.	Validation & Exception Blocks

3. Data Dictionary & Interface Schema

The following table outlines the foundational system parameters, required user inputs, and standard execution schemas used across both functional modules.

Data Field	Data Type	Description	Example Value(s)
IDX Stock Code	String	The ticker symbol representing a company listed on the Indonesia Stock Exchange.	BBCA, BMRI, ASII
Number of Lots	Integer	The volume of equity ownership held by the investor (1 Lot = 100 Shares).	10, 45, 100
Goal Amount	Float / Numeric	The target financial payout baseline specified by the user.	5000000, 100000000
Period	Enumeration	The time frequency constraint for checking or generating passive cash flows.	Monthly, Annually
Lifestyle Milestone	Enumeration	Preset regional expense configurations used to contextualize wealth thresholds.	Specialty Coffee, Gym Membership, Spotify
Annual Dividend (DPS)	Float / Numeric	Programmatically fetched real-time dividend per single share disbursed over the trailing 12 months.	225.00, 350.50

4. Feature Workflow & Core Mathematical Logic

The operational logic of Dividend Compass V2.0 is cleanly isolated into two standalone computational pipelines, guaranteeing precise functional delineation.

4.1 Feature 1: Lifestyle Calculator (Forward Feature Flow)

This engine translates an established equity portfolio into net passive income yield and projects real-world purchasing power against specific consumer milestones.

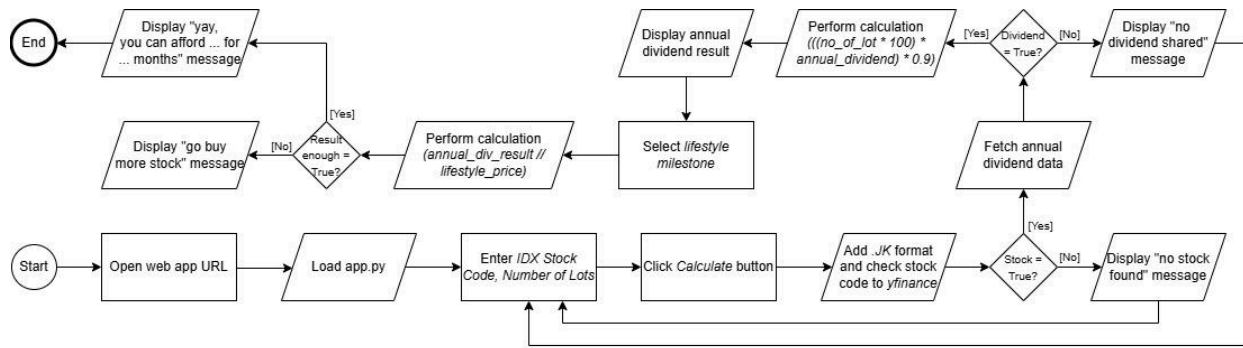


Figure 4.1.1 Lifestyle Calculator Flow

Mathematical Logic Execution Sequence:

- **Total Shares Realization:** Converts equity market lots into base raw share components ($\text{total_shares} = \text{total_lots} * 100$).
- **Gross Portfolio Income Yield:** Aggregates broad pre-tax earnings based on real-time DPS distribution tracking ($\text{gross_dividend_income} = \text{total_shares} * \text{annual_dividend_dps}$).
- **Net Income Calculation (10% Indonesian Final Tax Rule):** Deducts standard domestic fiscal withholding values ($\text{net_dividend_income} = \text{gross_dividend_income} * 0.9$).
- **Milestone Affordability Allocation:** Assesses overall lifestyle integration using an integer floor division loop ($\text{affordability_months} = \text{net_dividend_income} // \text{lifestyle_milestone_price}$).

4.2 Feature 2: Reverse Goal Planner (Backward Feature Flow)

This engine accepts a targeted net capital milestone and programmatically reverse-engineers the precise asset volume necessary to securely maintain that baseline.

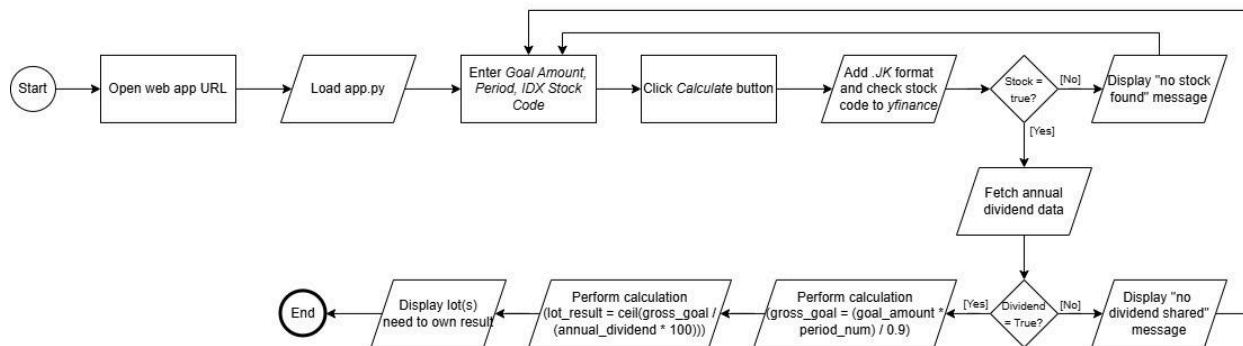


Figure 4.2.1 Reverse Goal Planner Flow

Mathematical Logic Execution Sequence:

- **Gross Revenue Target Adjustment:** Factors in withholding tax liabilities ahead of volume mapping ($\text{gross_target_goal} = (\text{goal_amount} * \text{period_multiplier}) / 0.9$).
Note: period_multiplier evaluates to 12 for 'Monthly' configurations and 1 for 'Annually'.
- **Lot Volume Requirement & Ceiling Execution:** Translates stock demands into

commercial market lot parameters. Utilizes an exact ceiling function to eliminate fractional asset shortfalls ($\text{required_lots} = \text{math.ceil}((\text{gross_target_goal} / \text{annual_dividend_dps}) / 100)$).

- **Boundary Constraint Implementation:** Ensures recommendations constantly sit above absolute transactional baselines ($\text{final_lot_requirement} = \max(1, \text{required_lots})$).
- **Capital Deployment Estimator:** Leverages active market validation close rates to determine overall liquidity funding expectations ($\text{estimated_capital_needed} = (\text{final_lot_requirement} * 100) * \text{current_stock_price}$).

5. Data Ingestion & Exception Management Framework

To deliver a professional-grade user experience, Dividend Compass implements a robust edge-case validation framework that maps directly to its architectural error-handling modules:

1. **Ticker Validation Failure:** If the data integration layer queries a stock code that does not exist on the Indonesia Stock Exchange, the exception block catches the network error, bypasses system halts, and renders a clean UI toast: "No stock found".
2. **Zero-Dividend Integrity Check:** In the event that a user selects a non-dividend-paying growth asset (such as specific technology tickers), the system triggers a validation block before division operations occur. This intercepts potential ZeroDivisionError faults, safely routing execution to display: "No dividend shared".
3. **State Isolation:** To preserve operational integrity as users toggle between options, all internal widget states are strictly decoupled using explicit, unique memory keys, preventing cross-tab variable pollution.

6. Future Roadmap & Success Metrics

6.1 Future Features & Enhancements (V3.0 Plan)

- **Automated Portfolio Sync:** Integration of automated PDF e-statement parsers to scrap active lot volumes seamlessly from local Indonesian brokerage apps.
- **Historical Yield Tracking:** Implementation of interactive visual line charts showcasing trailing 5-year dividend consistency and payout stability scores.

6.2 Key Performance Indicators (KPIs)

- **Calculation Velocity:** Reduce manual dividend yield calculation and net tax deduction computation time by >80% compared to typical spreadsheet workflows.
- **Tab Retention Rate:** Achieve >60% user interaction parity on the newly implemented Reverse Goal Planner module compared to the classic forward calculator.